AP PRECALCULUS

Unit 1 • Lesson 1.5 • Guided notes

Topic 1.5 • Polynomial Functions and Complex Zeros

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Connect real zeros, linear factors and x-intercepts.
- Use multiplicity to predict the graph near each zero.
- Apply the Fundamental Theorem of Algebra and the conjugate pairs rule; write polynomials from their zeros.
- Determine whether a function is even, odd, or neither.

Vocabulary. zero, factor, multiplicity, complex number, imaginary unit i, conjugate, even function, odd function.

1. Zeros, factors and intercepts

Key idea

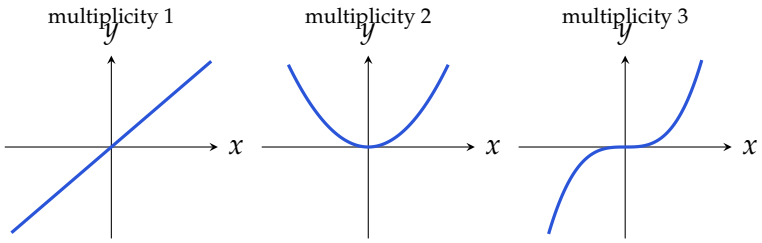
For a polynomial p and a real number a, these statements are equivalent: a is a _____ of p; $(x-a)$ is a _____ of $p(x)$; $(a, 0)$ is an _____ of the graph.

2. Multiplicity

If $(x-a)^k$ is a factor of $p(x)$ but $(x-a)^{k+1}$ is not, a is a zero of multiplicity ____.

Key idea

Even multiplicity: the graph _____.
Odd multiplicity: the graph _____; for multiplicity 3 or more it also _____ as it crosses.



3. Complex zeros

Key idea

Fundamental Theorem of Algebra: a polynomial of degree n has exactly ____ complex zeros, counted with multiplicity.
Conjugate pairs: if a polynomial with real coefficients has the zero $a + bi$, it also has the zero _____. Non-real zeros are _____ x-intercepts.

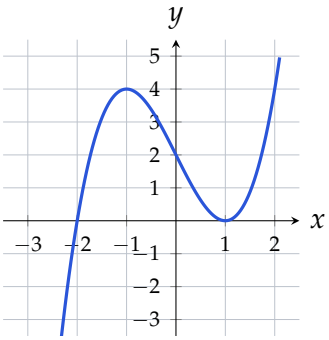
Example 1 • Finding all zeros

Find all zeros of $p(x) = x^4 - 5x^2 - 36$.

Example 2 • Building a polynomial

Write a polynomial of least degree with real coefficients, leading coefficient 1, and zeros -1 (multiplicity 2) and $2i$. Give its degree.

Example 3 • From a graph



The graph of a cubic q is shown. Write $q(x)$.

4. Even and odd functions

Key idea

f is **even** if $f(-x) =$ _____ for every x; its graph is symmetric about the _____.
f is **odd** if $f(-x) =$ _____; its graph is symmetric about the _____.

Example 4

Classify $f(x) = x^4 - 3x^2$, $g(x) = x^3 - 5x$, $h(x) = x^3 + 1$.

Common mistakes

- Listing non-real zeros as x-intercepts.
- Writing the factor for the zero -3 as $(x - 3)$.
- Forgetting the conjugate when building a polynomial with real coefficients.

Check your understanding

1. A degree-4 polynomial with real coefficients can have _____ real zeros (with multiplicity).
2. $f(x) = x^5 - 2x^3 + x$ is _____.